

Lecture Notes on

Quantum Mechanics

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These notes cover quantum mechanics at the undergraduate level. The sources used are:

- Griffiths: *Introduction to Quantum Mechanics*

Potential sources to consult in the future:

- Sakurai: *Modern Quantum Mechanics*

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1 The Schrödinger Equation

1.1 Wave Functions

Classically, we work with masses and attempt to solve for their equations of motions $x(t)$ using corresponding Newton's laws or energy equations. The quantum mechanics approach is different; the idea is to instead solve for a particle's wave function $\Psi(x, t)$ and from there we can do anything. The goal then is to solve the Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

After solving the Schrödinger equation for some initial wave function $\Psi(x, 0)$ the equation actually gives us all future values of the wave function $\Psi(x, t)$.

The statistical interpretation of the wave function goes as follows:

- Born's statistical interpretation of the wave function states that

$$\int_b^a |\Psi(x, t)|^2 dx$$

is the probability of finding the particle at $a \leq x \leq b$ at time t . Note that while Ψ is a complex valued function, $|\Psi|^2 = \Psi\Psi^*$ where Ψ^* is the complex conjugate of Ψ is always real and nonnegative.

- An obvious consequence is this is that even if you know everything about a particle (ie; you know its wavefunction), you still cannot predict with certainty the outcome of an experiment to measure its exact position.
- Before measurement, a particle's position is not well defined. Only after measurement can we say the particle has a "position," and by doing so we collapse the particle's wavefunction to a dirac delta centered around the measured position. This is known as the **Copenhagen Interpretation**
- The integral over all possible positions must be equal to one, since the particle must be located somewhere.

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

If we have a wavefunction, we need to make sure its normalized for it to be physical. We can normalize it to obey the integral by multiplying it by a scale factor. A glance at the Schrödinger equation one can see that if $\Psi(x, t)$ is a solution, then $A\Psi(x, t)$ is also a solution for nonzero constant A .

- One can also show that the normalization integral does not change with time, so there is no need to constantly renormalized solutions as the particle evolves with time.
- If we interpret $|\Psi(x, t)|^2$ to be the probability, then we can say the expectation value of position is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx$$

Note that the interpretation of this quantity is the expected value of the position from measuring particles with wave function Ψ . Simply remeasuring the same particle over again will not work, since the first measurement will collapse the particle into some other wave function with exact position

- By differentiating the expected position with respect to time, one can obtain the expectation value of the velocity. It's customary for us to work using momentum so we have

$$\langle p \rangle = -i\hbar \int \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx$$

- A more illustrative way to write the expectation values for position and momentum are:

$$\langle x \rangle = \int \Psi^* \cdot x \cdot \Psi dx$$

$$\langle p \rangle = -i\hbar \int \Psi^* \cdot \left(\frac{\partial}{\partial x} \right) \cdot \Psi dx$$

We say the x operator "represents" position and the $-i\hbar \frac{\partial}{\partial x}$ operator "represents" momentum.

- All classical quantities can be represented as a function of position and angular momentum. Then the representation of a classical quantity $Q(x, p)$ can be written as

$$\langle Q(x, p) \rangle = \int \Psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \Psi dx.$$

- **Ehrenfest's Theorem** states that expectation values obey their classical laws. For example, the time derivative of momentum is

$$\frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

1.2 Time Independent Schrödinger Equation

Let us assume that the potential $V(x)$ is independent of time. Then using separation of variables, we can substitute $\Psi(x, t) = \psi(x)\phi(t)$ into the Schrödinger equation giving us $i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V$. Then for the equality to hold, both sides must be equal to some constant value E leaving us with two ordinary differential equations:

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar} \phi$$

$$-\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V = E$$

The second equation is the **Time Independent Schrödinger Equation**. The first equation defines the characteristic time dependence and can be solved trivially $\phi(t) = e^{-iEt/\hbar}$. Separable solutions obey several important properties that we're interested in:

- These solutions are **stationary states**. What this means is while the wave function is obviously time dependent, $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, the probability density is not:

$$|\Psi(x, t)|^2 = \Psi\Psi^* = \psi^* e^{+iEt/\hbar} \psi e^{-iEt/\hbar} = |\psi(x)|^2$$

We can further show that the expectation value of any classical quantity is also not time dependent in a stationary state:

$$\langle Q(x, p) \rangle = \int \psi^* Q\left(x, -i\hbar \frac{\partial}{\partial x}\right) \psi dx.$$

- The constant E represents the Hamiltonian of the system. Classically, the Hamiltonian is $H(x, p) = \frac{p^2}{2m} + V(x)$ so the corresponding Hamiltonian operator is $\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$. So the time independent Schrödinger equation can be rewritten as

$$\hat{H}\psi = E\psi$$

Then the expectation values of the Hamiltonian is

$$\langle H \rangle = \int \psi^* \hat{H} \psi dx = \int \psi^* E \psi dx = E$$

$$\langle H^2 \rangle = \int \psi^* \hat{H}^2 \psi dx = \int \psi^* E (\hat{H} \psi) dx = \int \psi^* E^2 \psi dx = E^2$$

This means the standard deviation is 0. So any ensemble of stationary states with the same wavefunction have their Hamiltonian measured to be the exact same E .

- Every solution to the Schrödinger equation can be written in the form of a sum of stationary solutions.

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-E_n t/\hbar}$$

So we can basically decompose any wave function into the sum of wavefunctions associated with specific energy levels. The quantity $|c_n|^2$ represents the probability that a measurement of the energy will return value E_n , so we also must demand

$$\sum_{n=1}^{\infty} |c_n|^2 = 1$$

$$\langle H \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n$$

1.3 Infinite Potential Well

Consider the potential function given by

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & \text{otherwise} \end{cases}$$

Outside the potential well, $\psi(x)$ must be 0 because $V(x)$ goes to infinity. So we have $\psi(x) = 0$ for $x < 0$ and $x > a$. Inside the well, the Schrödinger equation gives us $-\frac{\hbar}{2m} \frac{d^2\psi}{dx^2} = E\psi$. The solution is just the simple harmonic oscillator with $k = \sqrt{2mE}/\hbar$. So the general solution is $\psi(x) = A\sin(kx) + B\cos(kx)$. Using the boundary condition that the wavefunction must go to 0 at $x = 0$ and $x = a$, we obtain

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

with corresponding energy

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

Some properties of the ψ_n functions:

- ψ_n alternates parity around the center of the well. ψ_1 , the ground state, is even, ψ_2 , the first excited state is odd, ψ_3 is even and so on.
- Each ψ_n are orthonormal. Namely:

$$\int \psi_m(x) * \psi_n(x) dx = \delta_{mn}$$

- They form a complete basis for any function $f(x)$:

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi}{a}x\right)$$

This is simply a fourier series for $f(x)$, and this can represent any function $f(x)$ by **Dirichlet's Theorem**. To actually determine the coefficients, multiply both sides by ψ_m^* and then integrate

$$\begin{aligned} \int \psi_m(x)^* f(x) dx &= \sum_{n=1}^{\infty} c_n \int \psi_m(x)^* \psi_n(x) dx = \sum_{n=1}^{\infty} c_n \delta_{nm} = c_m \\ c_n &= \int \psi_n(x)^* f(x) dx \end{aligned}$$

1.4 The Quantum Harmonic Oscillator

Consider the potential inside a harmonic oscillator $V(x) = \frac{1}{2}kx^2$. This potential is actually very illustrative because any general potential $V(x)$ can be taylor expanded into $V(x) \approx V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots$ but notice that we can simplify by working with x_0 as an equilibrium point x_0 , so the $V'(x_0)$ term drops, and we can also choose to define V up to a scalar constant such that $V(x_0) = 0$. If we assume $x - x_0$ stays small, then we could describe any oscillatory motion with small amplitude.

The Schrödinger equation is now

$$-\frac{\hbar}{2m} \frac{d^2\psi}{dx^2} = \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

where $\omega = \sqrt{k/m}$. To solve this we introduce the **ladder operators** $\hat{a}_-$ and $\hat{a}_+$

- We define the operator $\hat{a}_{\pm}$ as

$$\hat{a}_{\pm} \equiv \frac{1}{\sqrt{2\hbar m\omega}}(\mp i\hat{p} + m\omega x)$$

The product is $\hat{a}_-\hat{a}_+ = \frac{1}{2\hbar m\omega} [\hat{p}^2 + (m\omega x)^2] - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x})$. Here, $(\hat{x}\hat{p} - \hat{p}\hat{x}) \equiv [\hat{x}, \hat{p}]$ where $[\hat{x}, \hat{p}] = i\hbar$ is what's known as the **commutator** of $\hat{x}$ and $\hat{p}$.

- Using the ladder operators, notice that $[\hat{a}_-, \hat{a}_+] = 1$ and we can rewrite the Hamiltonian and Schrödinger equation as

$$\hat{H} = \hbar\omega \left(\hat{a}_+\hat{a}_- + \frac{1}{2} \right)$$

$$\hbar\omega \left(\hat{a}_{\pm}\hat{a}_{\mp} \pm \frac{1}{2} \right) \psi = E\psi$$

2 Dirac Notation

3 The Hydrogen Atom

4 Symmetries and Conservation Laws